

ORAL CANCER DETECTION USING RASPBERRY PI AND CNN FOR EARLY DIAGNOSIS

An AI and Computer Vision based preliminary screening system using Raspberry Pi

1. Project Overview

AI-Powered Oral Cancer Detection is an AI and computer-vision-based project developed to assist in the early screening of potentially abnormal oral conditions.

The system uses a Raspberry Pi and camera to capture images, processes them using OpenCV, and uses a CNN-based machine learning model to classify the images.

Goal: To explore an affordable, AI-assisted screening solution that can potentially support early detection.

2. How It Works

Camera → Image Processing → CNN Model → Classification Result

- Image is captured using the camera connected to Raspberry Pi.
- OpenCV is used for image processing.
- The processed image is given to the CNN model.
- The model analyzes the visual features and provides a classification result.

3. Technologies Used

Raspberry Pi · Python · OpenCV · CNN · Machine Learning · Computer Vision

4. Key Result

The developed model achieved approximately 78% classification accuracy during testing.

The project also involved research and patent-related work, giving practical exposure to developing AI solutions for healthcare applications.

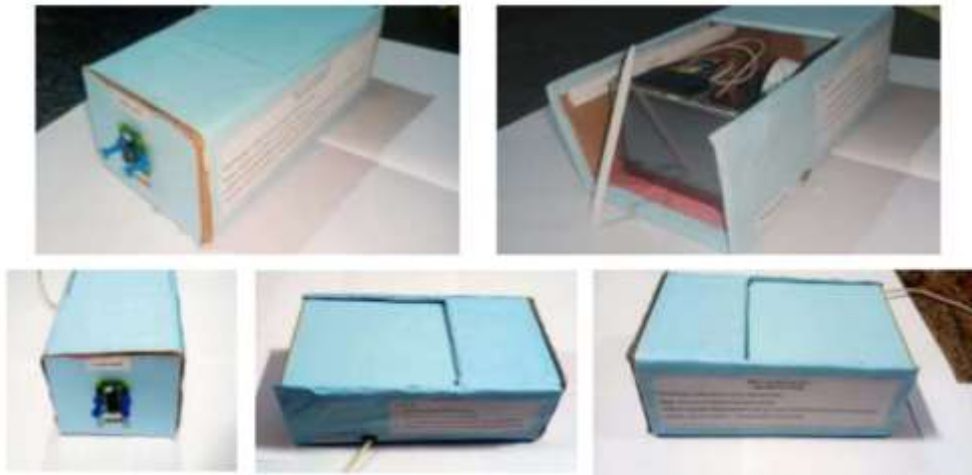
5. Applications

- Preliminary oral-health screening
- AI-assisted healthcare
- Remote/accessible screening concepts
- Edge AI and computer-vision applications

6. Future Scope

The system can be further improved using larger datasets, better AI models, improved image quality, and mobile/cloud integration.

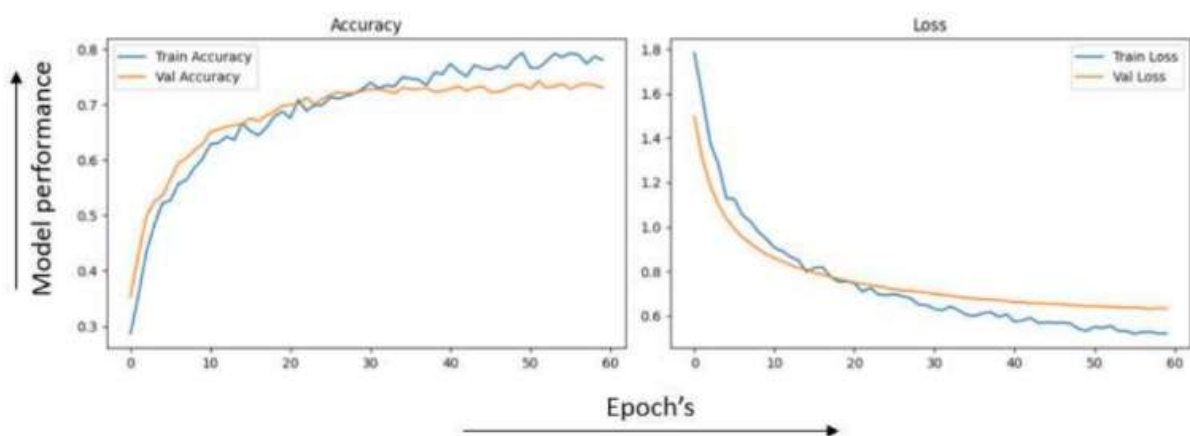
APPENDIX



Prototype images



Results



Model performance